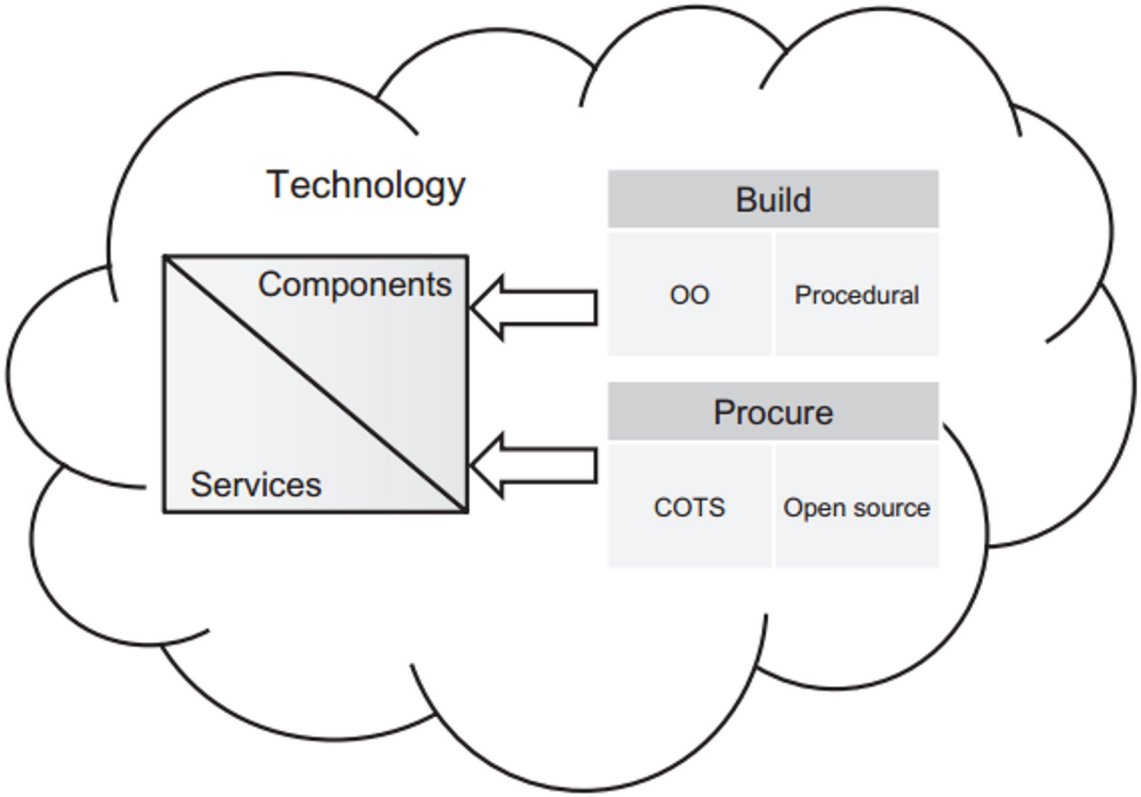
IF3141
SISTEM INFORMASI
Programming & Development Approaches

Semester I 2026/2027

Approaches to Development

- Very few systems are actually developed from the ground up
- Development teams are often merely assembling a series of ready-made components to form a solution to meet a particular set of requirements.

Figure 6.1 Development approaches: schematic overview



Build or Buy?

- For many systems development projects, perhaps the most fundamental decision to make early on in **the project is whether to procure a ready-made solution** or to build a bespoke (tailor-made) solution.
- Realistically, **the decision to build or buy is dependent on the nature of the project being undertake**n and, therefore, the type of system being sought.
- Hence, the decision on which approach to take will often not be made until the system requirements have been defined and agreed.
- A particular solution to a business need may be delivered through a **combination of bespoke developed and ready-made components**,

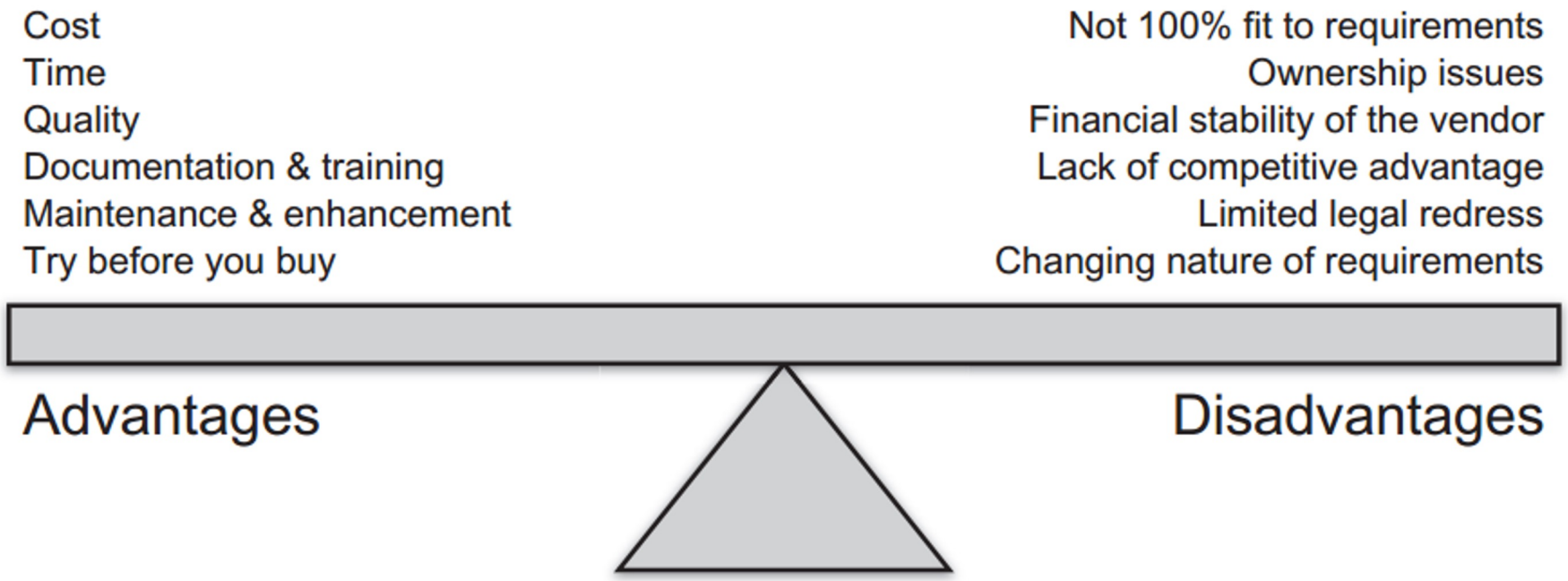
Build or Buy?

- **Bespoke**
  - At one time, bespoke systems development was the only option available to organizations **wishing to adopt computer technology**. In more recent times, it seems that only the largest and most wealthy of organizations have the luxury of dedicated in-house development teams, and even those organizations often outsource the development work to a third-party organization.
  - With the bespoke development approach, a ‘tailor-made’ solution is designed and developed to precisely meet a set of agreed requirements.
- **Ready-made**
  - There are two broad approaches to procuring a ready-made solution: purchase a **commercial off-the-shelf (COTS) solution** or obtain an open source solution.

Build or Buy?

- **COTS**
  - In the context of computer systems, COTS software products are ready-made software packages that are purchased, leased or licensed, off-the-shelf, by an acquirer who had no influence on its features and other qualities.

Figure 6.2 Advantages and disadvantages of the COTS approach



Build or Buy?

- **Open Source**
  - The customer obtains a ready-made solution but also gains access to the source code to enable in-house or third-party developers to tailor the solution to meet the precise needs of the customer.
  - The open source option clearly offers a number of benefits: zero license costs (although this doesn't mean a zero cost solution) and the provision of a ready-made solution that can also be tailored and maintained as necessary to suit the customer organization.

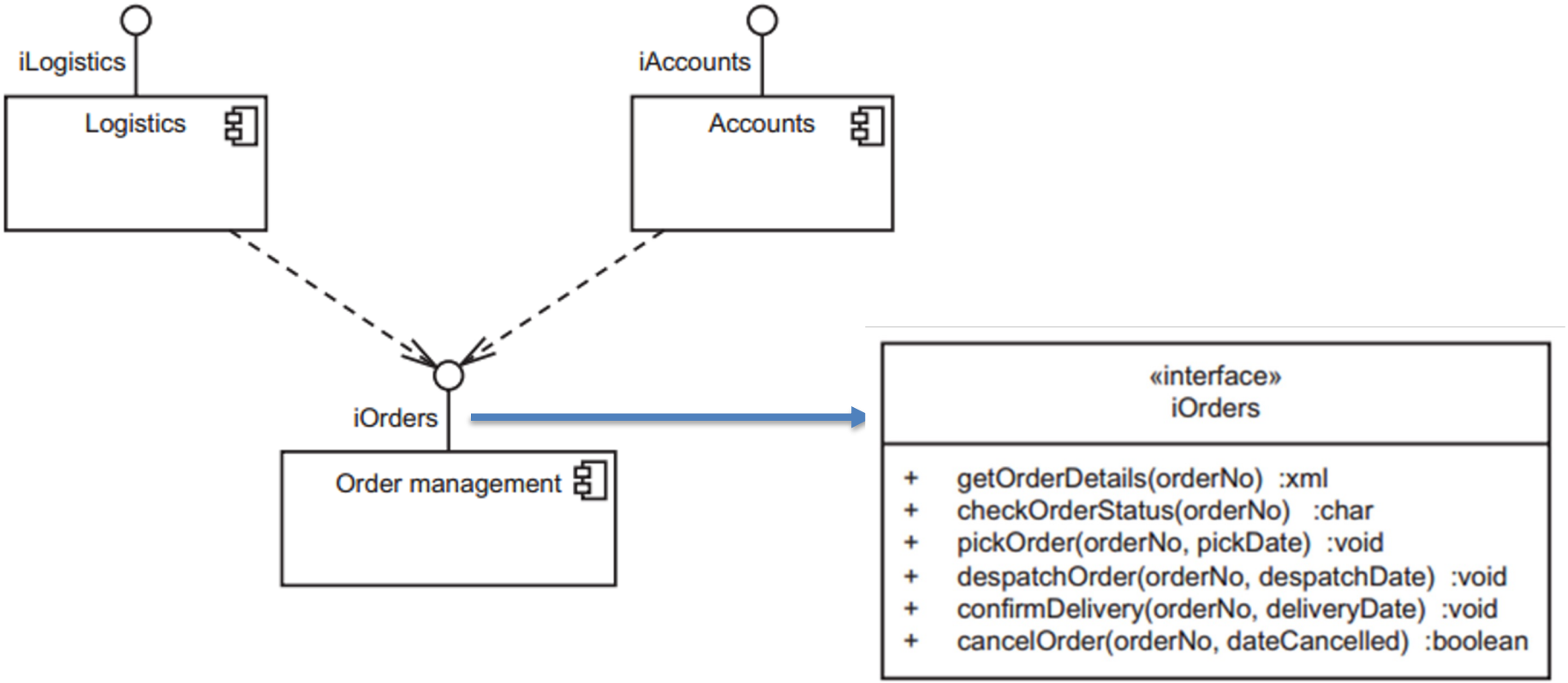
Component-based Development

- In practice, modern information systems comprise a series of separate, but integrated, building blocks referred to as components.
- **A component is something that can be deployed as a black box**. It has an external specification, which is independent of its internal mechanisms.
- With **component-based development,** software applications are assembled from components from a variety of sources; the components themselves may be written in several different programming languages and run on several different platforms.

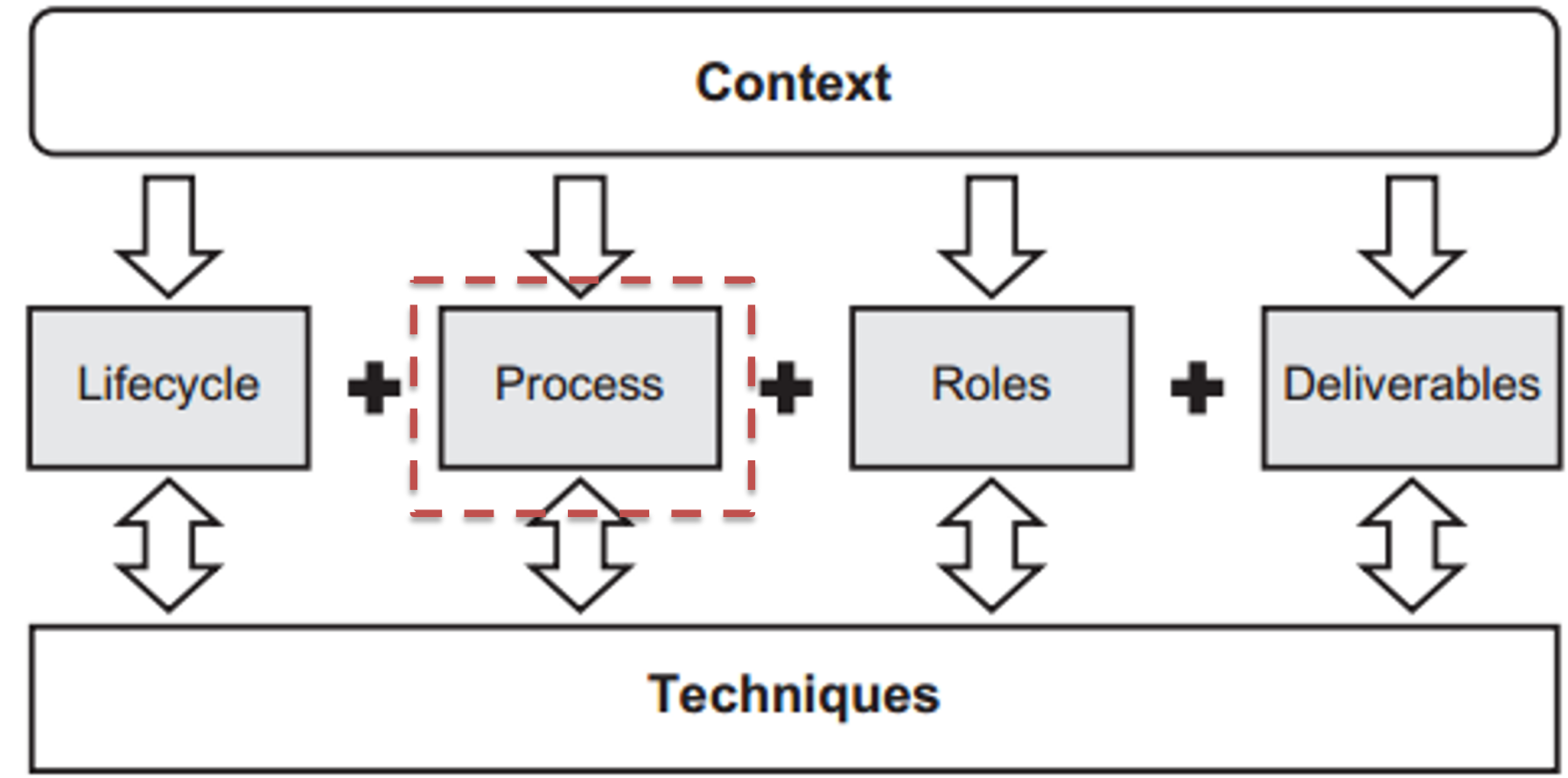
Component-based Development

- **Components can exist independently of each other** but can also be assembled together to build new solutions; they provide services to other components and use services provided by other components

Figure 6.3 Integrated components forming a Sales order processing solution



Development Methodologies



Development Methodologies

- Systems development processes broadly fall in to one of two categories: ‘defined’ or ‘empirical’.
- **Defined Process**
  - **The premise behind a defined process is that each piece of work can be completely understood** and that, given a well-defined set of inputs, the same outputs are generated every time.
  - Useful when the underlying mechanisms by which a process operates are reasonably well understood
  - Common defined (procedural) methodology and framework: SSADM, Structured Analysis, Structured Design, Yourdon Structured Method, Information Engineering, SADT, SSM, Structured Programming, JSP, IDEF0, Structured Methods, UP

Development Methodologies

- **Empirical Process**
  - The problem with applying defined processes to the discipline of systems development is that defined processes are based on the premise that **the product being built is always the same and, hence, the process is predictable and repeatable.**
  - In fact, systems development more closely resembles research and development of new products, which is more of an empirical process. Hence, the recent trend towards 'Agile' development, which is based upon an empirical process model.
  - **A process that cannot be repeated due to the inherent complexity needs to adopt this mechanism.**
  - Common empirical approach: **RAD, Agile** (XP, DSDM, Scrum, Adaptive software development, Crystal clear, Lean systems development, Test-driven development, Feature-driven development, Kanban, Agile UP)

Software Engineering Paradigms

- A number of different paradigms (world views underlying the theories and methodology of software engineering and programming) have evolved since the early 1960s, each proposing a radically different approach to building software.
- Three most prevalent paradigms that underpin modern software development:
  - **Structured programming**
  - **Object-oriented programming**
  - **Service-oriented development**

  Please explore it independently!

The Influence of Technological Advances

- **Advances in technology provide opportunities** for software developers that enable them to **deliver significant business benefits.**
- **Sometimes these benefits manifest themselves in cheaper and quicker solution development,** whilst others are more far-reaching and can actually revolutionize the provision of IT solutions for the entire IT industry.
- Two such technological advances are:
  - Cloud-based
  - Model Driven Architecture (MDA)

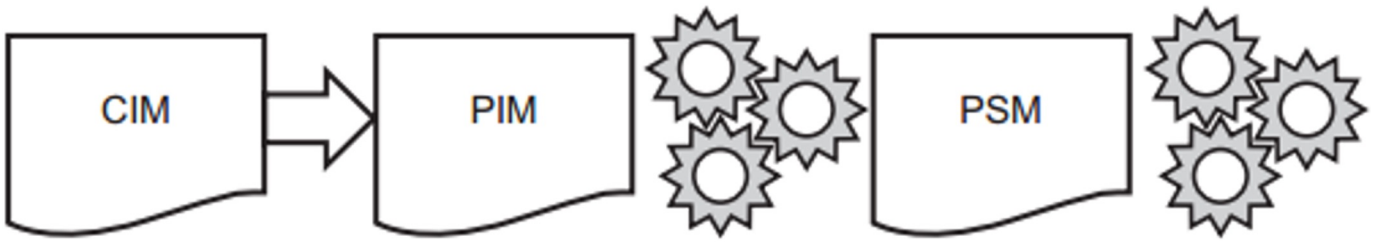
The Influence of Technological Advances

- **Cloud-based**
  - ‘a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources […] that can be rapidly provisioned and released with minimal management effort or service provider interaction.’ (NIST)
  - Cloud computing consists of three separate types of service:
    - **Software as a service (SaaS)** – the provision of applications in the cloud.
    - **Platform as a service (PaaS)** – the provision of services that enable the customer to deploy applications in the cloud using tools provided by the supplier.
    - **Infrastructure as a service (IaaS)** – the provision of computing power, storage and network capacity that enable the customer to run software (including operating systems and applications) in the cloud.

The Influence of Technological Advances

- **Model Driven Architecture (MDA)**
  - Model Driven Architecture refers to an approach to the **production of executable software directly from a set of design models**, without the intervention of a human programmer.
  - A major advantage of this approach, from the perspective of the development team, is that **the model and the physical system it represents are kept synchronized.**

Figure 6.7 The MDA process




```
/**
 * Simple HelloButton() method.
 * @version 1.0
 * @author john doe <doe.j@example.com>
 */
HelloButton()
{
 JButton hello = new JButton( "Hello, wor
 hello.addActionListener( new HelloBtnList

 // use the JFrame type until support for t
 // new component is finished
 JFrame frame = new JFrame( "Hello Button"
 Container pane = frame.getContentPane();
 pane.add( hello ) ;
 frame.pack() ;
 frame.show() ;                // display the fra
}
```